

Lipi — Technical Safety Brief

Clinical Extraction Pipeline: Architecture & Evidence

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Sprint 2 (Extraction Safety Hardening) · June 2026

Core Safety Claim: Every clinical fact Lipi produces is either a verbatim span from the transcript or a canonical medical alias of one. Lipi never generates, infers, or fabricates a clinical value.

This is not a policy — it is an architectural constraint enforced at two layers with an automated test suite (103 tests) and an eval harness with safety-critical golden cases.

1. Why Generative Models Are Excluded

Large language models (GPT, Gemini, Llama) are excluded from Lipi's extraction path for one reason: **they cannot guarantee transcript-grounding**. In clinical documentation, a fabricated fact that enters a medical record can cause drug interactions, wrong treatments, or billing fraud. The probability of hallucination does not need to be high — it needs to be zero.

2. Pipeline Architecture

```
Transcript (Hindi / Hinglish / English)
  |
  v
[LAYER 1 --- Deterministic Extraction]    clinical_extractor.py
  * Bilingual keyword lexicon (EN + HI + Hinglish transliteration)
  * spaCy sentence boundary detection
  * Negation-aware regex guards (40-char pre / 25-char post window)
  * Regex vitals: temperature, BP, SpO2, blood glucose, weight
  * Diagnosis: exact lexicon match only --- no fuzzy matching
  * Context guard: "known case of X" / "on treatment for X" != new Dx
  |
  v facts: symptoms, vitals, medications, investigations,
    allergies, follow_up, diagnoses
  |
[LAYER 2 --- GLiNER Augmentation]          clinical_pipeline.py
Model: urchade/gliner_medium-v2.1 (extractive NER)
Fields: medications, investigations, allergies, follow_up ONLY

EXTRACTIVE SAFETY GATE (runs before merge):
for field, items in gliner_facts.items():
    for item in items:
        if item.lower() not in transcript.lower():
            logger.warning("extractive-safety violation: %r", item)
            # item is discarded
```

```

GLiNER cannot add: symptoms, diagnoses
|
v
Merged facts -> SOAP generator -> CDS alerts -> API response

```

3. Field-Level Safety Policy

Field	Method	Safety rule
Symptoms	Lexicon + transliteration	Exact or canonical alias. Negation-aware.
Diagnoses	Exact lexicon only	No fuzzy. No inference. Context guard suppresses "known case of X" from populating diagnoses. Empty if doctor did not state one.
Vitals	Regex only	A model may never emit a numeric value.
Medications	Regex + GLiNER	GLiNER output verified verbatim in transcript before merge.
Investigations	Keyword + GLiNER	Same extractive gate.
Allergies	Pattern + GLiNER	Extractive gate. Penicillin stays in allergies, not medications.
Follow-up	Pattern match	Decimal-aware: "if fever above 39.5" captured in full (not truncated to 39).

4. Sprint 2 Safety Fixes

4.1 Diabetes context leak (*high severity*)

“Known case of diabetes on metformin” populated the diagnoses field via a "sugar" -> "Diabetes Mellitus" keyword mapping. Fix: removed the ambiguous keyword; added `_is_context_mention()` — a pre/post context window guard detecting medication-context language (“for X”, “on treatment for X”, “known case of X”).

4.2 Vitals regex gaps (*medium severity*)

Three failures: (a) “38.5 **degrees** Celsius” (plural) — fixed by `(?:degrees?s*)?`; (b) “BP 140 **by** 90” — added **by** as separator alongside **/** and **over**; (c) “fasting sugar **today is** 160” — added `[^.;]{0,20}?` to allow intervening words.

4.3 Follow-up decimal truncation (*medium severity*)

Pattern `[^\.\n;.,]+` stopped at the `.` in “39.5”, capturing “39” only. Fix: `(?:[^\n;.,]|\.(?=\d))+` — a period is allowed only when immediately followed by a digit.

4.4 GLiNER extractive gate (*critical, new control*)

No runtime verification existed that GLiNER outputs were actual transcript spans. Added a post-inference check logging a WARNING for every output not found verbatim in the transcript. Gate runs on raw GLiNER output before merge, so Layer 1 canonical names (e.g. “Fasting Blood Glucose”) are not checked — only GLiNER’s own output is.

5. Evaluation Evidence

Test suite: 103 unit tests, all passing.

Test class	Tests	Coverage
TestSprint2DiabetesContextLeak	5	Diabetes context guard
TestSprint2VitalsRegex	5	Temperature plural, BP "by", glucose
TestSprint2FollowupTruncation	4	Decimal preservation
TestSprint2GLiNERSafety	5	Extractive safety gate
Prior regression tests	84	Full extraction baseline
Total	103	All passing

Eval harness: 6 golden cases, F1 = 1.00 on all 10 fields.

Case	F1	What it tests
T1	1.00	Basic English OPD
T2	1.00	Hindi/Hinglish (bukhar→fever, teen din baad)
T3	1.00	Safety: diabetes context → diagnoses must be empty
T4	1.00	Vitals: degrees plural, "by" separator, glucose
T5	1.00	Follow-up decimal: "if fever above 39.5"
T6	1.00	Safety: penicillin in allergies only; steam not a medication

Cases T3 and T6 are safety regressions — a failure there means patient-harming output, not a metric drop.

6. Hard Prohibitions

The following constraints are enforced by architecture and are not configurable:

- No generative/LLM model in the clinical extraction path (no GPT, no Gemini, no Ollama)
- No symptom inference (“patient likely has X”)
- No fuzzy diagnosis matching
- No free-text capture as a diagnosis without explicit assessment language
- GLiNER cannot add symptoms or diagnoses
- A numeric vital value may never originate from a model

7. Regulatory Position

Lipi is a **documentation assistant**, not a diagnostic device. It does not diagnose, prescribe, or recommend treatment. All clinical decisions remain with the licensed physician. Physician review and digital co-signature are required before any note enters a medical record.